Improving a CTO System between Sub-Assemblies and the Final Assembly Line through Value Stream Mapping

"A Case Study of Lancer Corporation's Production Department"

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ABSTRACT

Lancer Corporation prides itself in being an industry leader of beverage dispensing equipment for over 50 years and has grown substantially within the past five decades. However, with such high demand and a large variety of Ice Beverage Dispensers and patented LEV valves, eliminating waste and creating a lean manufacturing environment has never been more important to Lancer Corporation and their large customer base. The objective of this project is to observe and evaluate the current state through value stream mapping (VSM) of the CTO system between Sub-Assemblies and the final assembly line. The current productivity rate is about 20 to 24 units a day and Lancer's target is to produce 32 units a day to fulfill customer demands. Through determining the value added and nonvalue added steps which are required to produce one unit, an improved value stream map will be proposed to eliminate process wastes and increase efficiency. All suggested equipment and tools included in the Future State Map vision will be discussed and overall lessons learnt are summarized.

Keywords

Lean, Value Stream Mapping, sub-assembly, final assembly, cycle time, takt time

INTRODUCTION AND BACKGROUND

Lancer Corporation is a leading manufacturer and marketer of beverage dispensing systems and strives to consistently design, manufacture, and market superior quality products. Lancer is considered a vertically integrated manufacturer, meaning they control their supply chain from manufacturing to end sales. While this gives them a strategic advantage within the beverage dispenser industry, it also creates that much more pressure to maintain a lean environment; they must keep the plant running and maintain efficiency for large profit margins. Additionally, the variety of dispensers being produced at Lancer further complicates their manufacturing. Lancer currently manufactures a complete range of mechanically cooled and ice cooled soft drink dispensers, dispensing valves, carbonators, beer dispensing equipment, and an extensive line of beverage dispensing parts and accessories [1]. Lancer Corporation has realized the importance of lean implementation and has completed many projects revolving around lean principles to improve their facilities. For purpose of scale, this paper focuses on the manufacturing facility headquarters in San Antonio, Texas and, more specifically, the CTO system between sub-assemblies and the final assembly line for eight dispenser products families. From the Flavor Select models with a 22'' footprint, to the Sensation, which includes 8 outlets and is 30'' wide, the subassembly line includes all parts required for each model. An image of a popular dispenser manufactured along this line is shown in Figure 1 [1].

Figure 1 Lancer Corporation's Flavor Select 22 Dispenser Model

As orders are received from the customer, each station within the subassembly line works to gather the hardware necessary for that specific model. Following into the final assembly line, the dispenser is put together and sent out for shipment to the consumer. With an average cycle time for each dispenser model that is not meeting customer demand, there is room for improvement. Through implementation of lean principles and concepts, waste can be eliminated, and the cycle time reduced. The manufacturing processes for two models (FS 22 shown above and Mercury 30) were observed at Lancer Corporation's headquarters and cycle times were recorded for each individual process. This data was used in congruence with lean tools and principles, to develop a future vision that eliminates waste.

LEAN PRINCIPLES

Lean manufacturing aims to eliminate waste and continuously strive for perfection. While efficient flow should be the natural state in any manufacturing environment, this is often not the case. Flow is defined as the continuous movement of material through the manufacturing processes and on to the customer [2]. Achieving continuous flow within the manufacturing process supports companies in achieving better quality products, utilizing less floor space, having a quicker response to problems, and maintaining lower inventories, to name a few benefits. However, when flow is not continuous and there are a lot of start/stop processes, waste is created and the environment veers from the ideal lean manufacturing vision. There are seven wastes addressed in lean manufacturing; waiting, over-processing, motion, overproduction, defects, inventory, and inefficiency. Additionally, underutilization of human talent is often considered an eighth waste; for example, operators along the assembly line are not sufficiently trained to perform different work functions. Understanding and identifying waste within the process is the first step toward a lean environment. In order to identify the waste, Value Stream Mapping is used to examine and record all of the activities along the value stream as the product is transformed from raw material to a finished product. Each action taken to produce the final product is analyzed to find room for improvement. This process alone will help bring waste to the surface so it is apparent where improvements can be made because it will be evident which steps add value and where efforts are being wasted. After mapping the current state, a future state is established in which the goal is to eliminate waste and create a continuous flow and a plan is developed to achieve that vision. While working toward a lean environment, it is important to always keep in mind the principles of lean manufacturing to ensure success. The first principle is that the customer defines the value; this means that the engineer must realize what the customer is willing to pay for and therefore

determine within each process what is value-added compared to non-value-added. This principle plays the most important role in eliminating waste. Absolute efficiency involves the second principle is that the customer establishes pull for the production. This principle emphasizes that each process should only make what is needed by the customer, when the customer wants it, and in the proper amount [2]. A pull system will make the final production the same as the customer's consumption rate and will essentially eliminate the waste of inventory. The third principle in lean manufacturing is to empower the people adding value; it is important to have empowered, synergistic team members as they are on the front line of implementation. The fourth principle is to eliminate waste in the Value Stream; as previously discussed, this is achieved by mapping the current state and working toward a future state vision through waste elimination. The fifth and final principle is to use total system cost to drive performance. Keeping this principle in mind will increase quality and safety, and decrease response time because it will minimize total system per unit cost. While these five lean principles may be difficult to implement, especially into an existing system, the benefits will ultimately outweigh the difficulties as efficiencies and profitability are increased.

CURRENT PROCESS: SUBASSEMBLY AND FINAL ASSEMBLY LINES

Lancer Corporation's headquarters in East San Antonio manufactures a large variety of products from their large frozen yogurt dispensers to their own customized valves. The focus area for this project was the CTO System in which they fulfill customer demand for 8 different soda fountain machine dispenser product families. This line consists of two subassembly lines and one final assembly line. Shown below in Figure 2 is the subassembly line in which 7 stations are lined up parallel to one another. Lancer Corporation's CTO system between their subassembly and final

assembly line utilizes one-piece flow. One-piece flow focuses on completing the entire production of one product while minimizing work in process inventory.

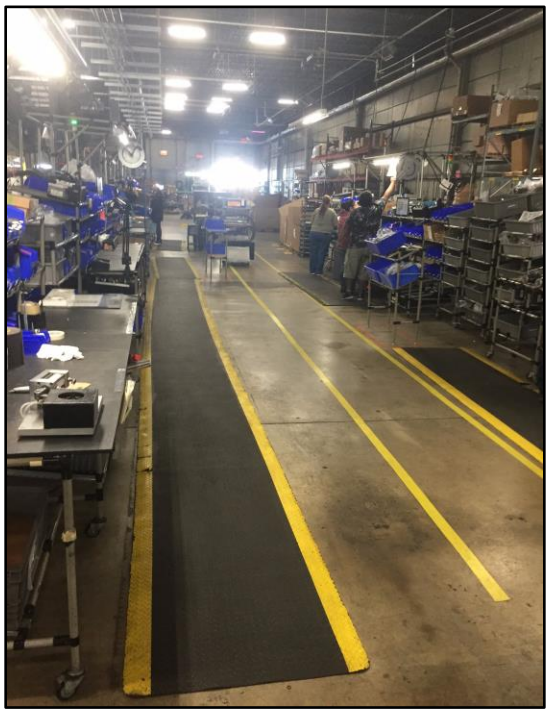

Figure 2 Subassembly Line

Lancer's assembly line has one cart for each product that is moving along this cart collects all subassemblies required for that unit. Each cart is labeled with the correct model name and part number as shown in Figure 3 below.

Figure 3 Subassembly Cart for Mercury 30

Following the collection of all subassemblies, the cart is pushed alongside a base unit through the final assembly line. This final assembly line pictured in Figure 4 performs the installation of all subassembly parts collected as well as testing final cosmetic touches.

Figure 4 Final Assembly Line

Gemba is a Japanese term which means “the real place”, in lean it is the place where value is created. The first step before beginning any lean initiative would be to go on a gemba walk and

observe each process as it is happening. Seeing first-hand how things are done on the floor, will prove extremely valuable when trying to identify and eliminate waste. A more detailed discussion of each individual substation along the subassembly line, as well as observations seen during site visits are included in the subsequent section.

CURRENT STATE

Before the stream mapping of CTO between subassemblies and the final assembly line was created, there was a need to read product work instructions to understand fabrication process and required tools for each assembly. Unfortunately, there is no work instructions or any documentation regarding CTO processes, and all manufacturing processes are performed based on operator's experience. In addition, an extra operator is required to work in each assembly station to train a new operator and ensure subassemblies are constructed correctly, which causes increase in product's assembly cost. According to the Production Manager, Tatyana Sanchez, the current cycle time is 20 minutes per unit and productivity rate range is 20 to 24 units per day. Although, during our observation for the process throughout two days, we determined the average cycle time was 30 minutes per unit. This Value Stream Mapping project concentrated on the material and information flow through all subassembly processes up to the packaging process. The design of all subassembly stations are similar as shown in Figure 2 and each have parts on stacked shelves at eye-level of the operator, as well as on both sides of the table, resulting in a long subassembly line and a long distance between each station. A CTO cell team lead provides information flow regarding required subassemblies to each subassembly station for the entire day. The basic concept of flowing the subassemblies through the process is to use a cart as shown in Figure 3, which includes a product name, part number, and an order sheet. There are 12 carts flowing through the

subassembly to final assembly process. The process was observed for the FS 22 product shown in Figure 1. The subassembly process consists of seven subassembly stations. An operator places a subassembly on the cart and signs in the order sheet to record the process. And then the cart is pushed to the next subassembly station until it is filled with all required subassemblies and then moved into the final assembly process. The process A (Kitting) consists of making a kitting bag, walking to the two supermarkets to get splash plates and other parts, assembling the plastic pipe-foam tube using a specialized machine, and placing each of these subassemblies on the cart. The process B (Power Box) is the assembly of the power box and fluorescent lighting if applicable, the operator will also gather the touch pad and nozzles assemblies from process C and placing all items on the cart. The process C (MVU) does touchpad assembly and gets the nozzles assemblies from inventory and subassemblies are push into process B, as mentioned earlier. Process D does batching of nozzles for unlimited orders and these nozzles are stocked between process C & D. Also, it does batching of nozzle components to facilitate the assembly of nozzles including cutting hoses, placing O-ring on fitting, installing and crimping hose-fittings. Furthermore, the time spent to assembly nozzle varies from a product to another based on number of outlets. The process E (merchandise sticker process) is simply to get a plastic cover from the supermarket, which is next to the machine, and a required sticker is placed on it as shown in Figure 5, and then an operator will set the subassembly on the cart. Process F is an assembly of valves, which are assembled in another cell and placed in supermarket behind processes B & C, and some adjustments are to be done on the valves based on the product family.

Figure 5 Merchandise Sticker

The cart is then pushed into process G. Process G consists assembly of motor, ice chute, solenoid, and wrappers preparation. After all, the cart is pushed into the final assembly line. The beverage dispenser base is brought to the final assembly process by a warehouse operator after it is checked by Quality Dept. The final assembly line consists of four assembly processes and two testing stations, as shown in Figure 6. A base dispenser machine is placed on bullet and the height of the base dispenser is adjustable by using a pallet jack. The 1st final assembly process, which includes installation of the ice chute, solenoid, removing wrappers from panels, and installing the side bracket, takes about 16 minutes. The 2nd final assembly process, which includes installing trim, valves, nozzles, the power box, and wire connection, takes about 7 minutes. The 3rd final assembly process includes touchpad assembly and takes 6 minutes. The testing 1 process takes about 6 minutes and includes programming, ice chute and valves testing, and a Hipot test. After the first testing is over, a unit is moved to the second station of testing. There are three stations for the second testing process and, if all are occupied, then the unit is placed aside waiting for the testing. The second testing includes flow, pressure, and sanitation tests and takes about 20 to 45 minutes based on the size of unit and number of nozzles used in assembly. Sometimes testing 2 takes longer

than mentioned above due to failing in the test and troubleshooting by QC is required to investigate and eliminate the cause. During the observation process, there were three units waiting for the second testing. The last assembly process, which includes merchandize sticker cover installation, connection check, panels placement, plastic pipe-foam tube installation, and lighting installation if applicable, and takes about 15 to 25 minutes. And then the finished unit is pushed into Warehouse for the packaging process. During the observation, there were four units waiting for final assembly.

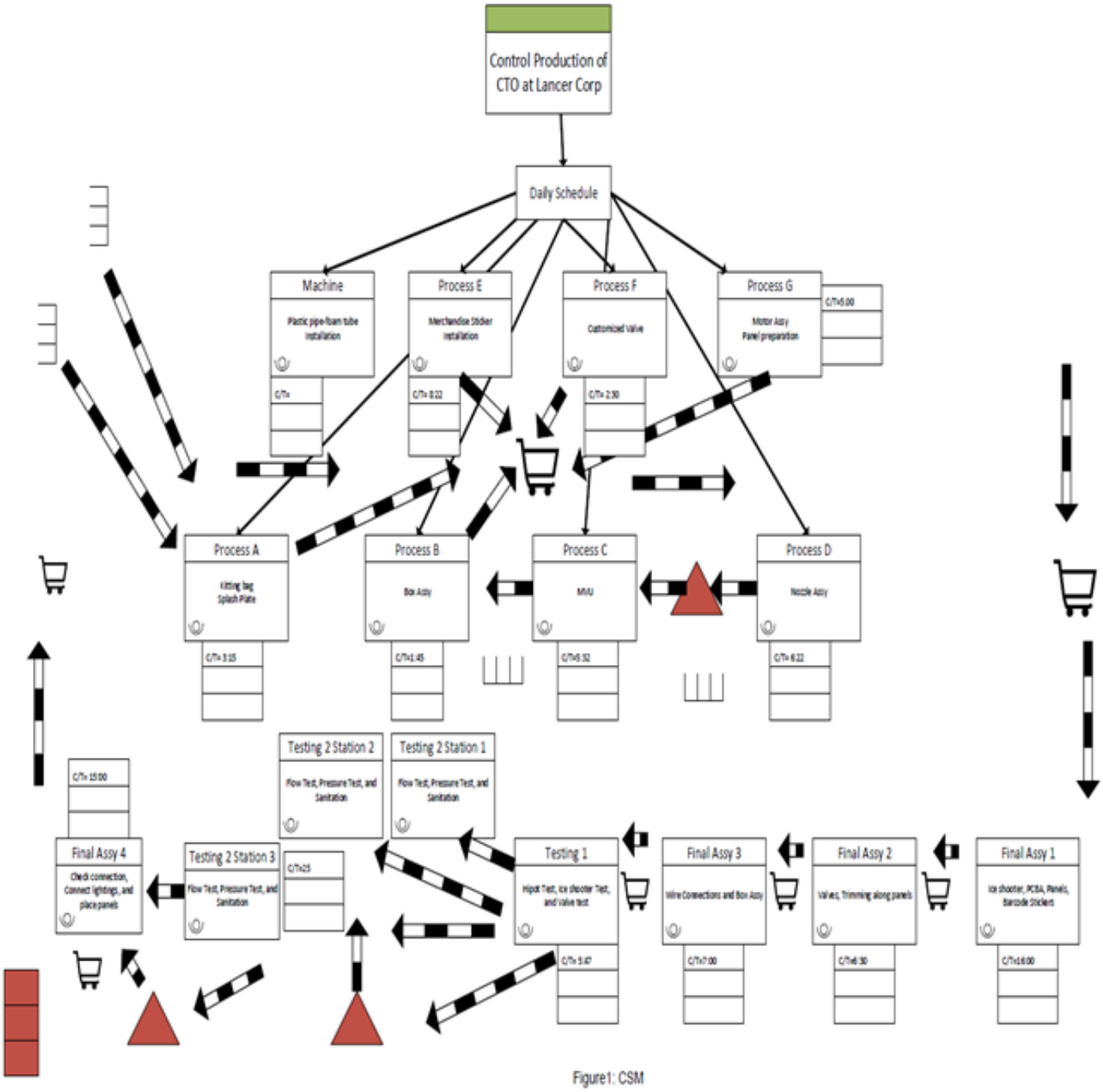

Figure1: CSM

Figure 6 Current State Map (CSM)

EVALUATION OF CURRENT STATE

After mapping the current state of assembly flow in CTO, it became visible why there are flow process and batching problems. The major issue is that there are no work instructions or process sheets for each product family as a reference to ensure doing the work properly or helping to train new operators. Most of the main operators have been working for the same assembly process for a long time and they ultimately depend on their experience to perform the job. Therefore, the product could possibly be assembled incorrectly, resulting in increased defects rate. Therefore, training new operators process takes a long period of time in which an experienced operator needs to watch a new operator doing the work for several days to ensure they can assemble all product families properly, resulting in increased product cost. The times of each subassembly process is imbalanced due to required motion to get parts from one or more supermarkets for some processes, allowed batching, and some processes taking longer than others. Process A's operator needs to walk to supermarkets in different locations to get needed parts and to assemble plastic pipe-foam tube by using the machine to get the hose ready. In addition, some subassembly stations are too long compared with needed space to perform the work for example nozzles are moved upstream from process D to B; however, nozzles are not used in either process C or process B. The first, second, and third final assembly processes are imbalanced whereas the first assembly process time is more than the second and third processes together which causes slacking in final assembly process 2 & 3 as well as wasting time. Therefore, imbalanced final assembly processes typically cause more than two units waiting either for testing or the fourth final process. In addition, the imbalances between the subassembly and final assembly processes cause either several ready carts waiting for the final assembly or no available carts to continue the subassembly flowing. Therefore, this issue results in stop production in subassembly for a half hour or more until the

subassembly line gets the cart to resume operation. During the non-running operation period, only three batched subassemblies are allowed. This situation results in that several operators do not have work to do.

SUGGESTED FUTURE STATE MAP

When analyzing the current state map, waste was discovered. Unnecessary motion, waiting, inventory, and underutilization of human talent were all observed along the subassembly and final assembly line. As seen in Figure 7, a Future State Map has been developed with the goal of eliminating these wastes. The first issue addressed when planning a vision for the future state, was to analyze the cycle times of each process and find a solution to balance these times for continuous flow; this required significant re-organization of the production line. One issue previously mentioned is the second testing station, in which the dispenser is tested for proper flow and pressure and then sanitized. The problem with this process is that it bottlenecks the final assembly line, as it takes anywhere from 30-90 minutes to complete based on the model and possible problems the operator might encounter. To mitigate this problem, it is suggested that there are two separate final assembly lines. This would be beneficial due to the variety of models; with a final assembly line dedicated to the larger models and one specifically for the smaller ones, a bottleneck caused by testing the larger models will be eliminated. An order for a small to medium sized dispenser will move through the line without delay at the testing station. Consequently, this new setup is also a better utilization of space, as the subassembly processes become one line rather than two. Further, to balance the subassembly line, the processes were re-organized such that they are aligned to have a balanced cycle time with their respective final assembly process.

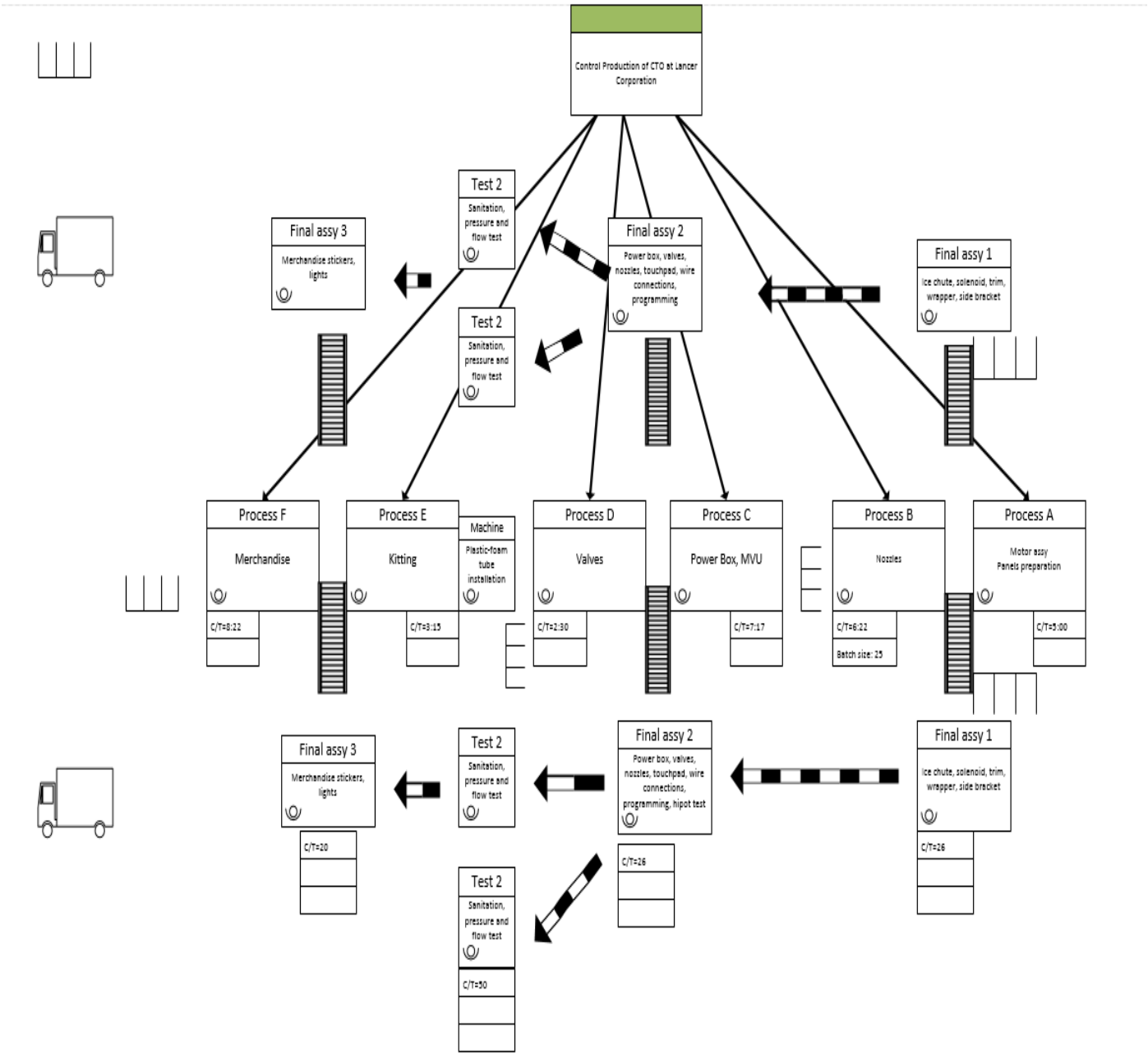

Figure 7 Suggested Future State Map (FSM)

To determine the best setup, processes were evaluated based on cycle time and final assembly contribution. The first three stations along the final assembly line were not balanced with process times of 16 minutes, 6.5 minutes, and 7 minutes, respectively. To balance these times, each process

was carefully looked at in order to identify which subassembly installation may be moved to a separate station. After deliberation, it was decided that these three stations would be decreased down to two; one station will install the ice chute, solenoid, trim, and side brackets while the other will take care of valve, power box, nozzle, and touchpad installation. With this reduction of stations, the line is made more compact and organized, as seen when comparing the CSM to the FSM. These changes also brought about the realization that the Power Box and MVU processes may be combined to form a single station along the subassembly line. These processes were previously working together but at separate stations, which caused unnecessary motion and wasted space. As seen in the Future State Map, these processes were combined into Process C. The order in which the subassembly stations were in did not previously matter, as each subassembly went to the same location before moving to the next step. With the implementation of conveyor belts, this flexibility will no longer be present. This forced a careful consideration of where to place each process block along the line. It was ultimately decided that these process blocks would be placed based upon final assembly setup. This allowed for further cycle time balancing and a more efficient setup. Placing the processes strategically along the line will allow less down time for later processes. This brings up the next problem that was focused on, which is unnecessary motion. The current state, as discussed, uses a cart to gather all subassemblies and is then moved through the final assembly line. Conveyer belts are proposed as an addition to the line to eliminate this cart movement. These conveyors will allow the operator to remain at their station while transferring their subassembly directly to the respective final assembly station. Another source of unnecessary motion is the inconvenient locations of supermarkets in the warehouse. For example, the supermarket used by the kitting process requires the operator to walk to two separate locations to gather supplies. The future state map suggests these supermarkets are moved to the station that

uses them to eliminate this waste of motion. These changes are shown in the Future State Map next to Process F and Process B and C. After reducing waiting time and unnecessary motion, possible solutions to the waste of inventory were suggested. In Process B where nozzles are made, there is a large inventory because the operators work at a fast pace. It is suggested in the future state that the operators limit production to batch sizes of 25 nozzles. This will provide enough nozzles to manufacture the next six dispensers. This solution also addresses the next waste, which is underutilization of human talent. It is recommended that all operators are trained throughout the line so down time at one station will allow those operators to help on another process. For example, when the operator at the nozzle station is finished with their batch, they would be utilized where help is needed. While these implementations will greatly reduce waste along the production line, it is still necessary to focus on continuous improvement and strive for perfection; this is an important aspect of lean manufacturing and it is recommended that these changes are applied and a follow-up re-evaluation is performed for further improvement. Lean tools and the implementation of equipment will be further discussed in the next section.

SUGGESTED TOOLS/EQUIPMENT

As mentioned in the suggested future state map, conveyors will be set up between subassembly stations and vertically to two final assembly lines to push subassemblies into two directions. Therefore, the cart will no longer be needed, which reduces an unnecessary motion and eliminates non value-added process, as well reducing utilized floor space. Currently, Lancer Corp. has conveyors in their possession, but the type of conveyor is unknown. The suggested conveyor is a belt conveyor as shown in Figure 8. This conveyor was chosen to protect subassemblies from

scratches and to prevent small subassemblies from falling off as they might if roller conveyor was used. Four 6' Overall Length x 12" W, fixed speed belt conveyors will cost around $ 8,720.

Figure 8 Belt Conveyor

In addition, a quality concern was observed during the nozzle process in which the operators spent a large amount of time cutting the tubing used for each subassembly. An operator spend a significant amount of time cutting each tube into the required length. Another concern is that the tubing often gets damaged or pinched during the cutting process as shown in Figure 9, and these damaged pieces may be accidently pass to the next operation in the nozzle assembly.

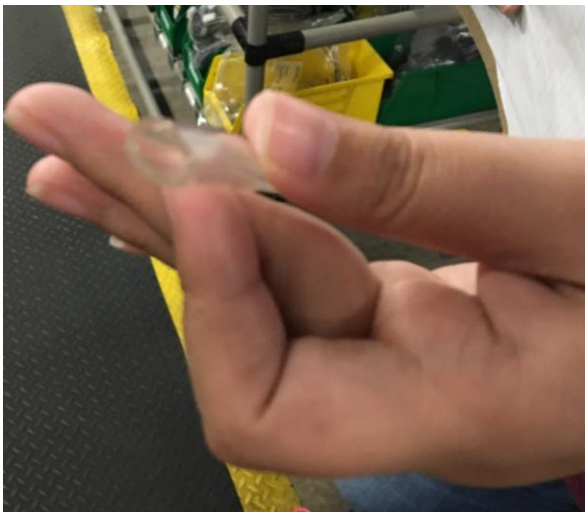

Figure 9 Damaged Tube due to Manual Cutting Process

This equipment will be beneficial to reduce defects per million opportunities (DPMO), increase efficiency of the nozzle assembly process, reduce number of operators required for the process, and increase safety in the process. This machine, as shown in Figure 10, provides a precise length and reduces the number of defects. An operator only needs to setup the machine and enter required length and quantity of hoses to be cut.

Figure 10 Automatic Tube Cutting Machine

Another suggested piece of equipment, that will likely increase productivity of the nozzle subassembly process, is a pneumatic hose clamp instead of manual pinch-on clamp tool. This machine as shown in Fig 11 will be useful to ensure that the fitting-hose clamp is fully secured, reduced broken clamp during the process, and facilitate nozzle subassembly process. The two final assembly lines will be able to run one or two different products at the same time, in the case there are prompt orders. In order to perform at that capacity, an error proofing method should be used. A suggested method to prevent mistakes is to place a label on each subassembly prior to moving it down the conveyor. The label should include a product name and number, and could be a different color to easily distinguish which model a subassembly belongs to, and whether or not it should be pushed to the final assembly line. Therefore, this method will prevent the possibility of

using the incorrect subassembly on the final product assembly because the subassembly will be checked by two operators.

Figure 11 Pneumatic Hose Clamping Tool

POSSIBLE CHALLENGES OF IMPLEMENTATION

One of the challenges to implement new production process is down time of assembly line to setup new equipment as well as execution of new assembly station layout. To perform this duty without stopping production could be done by setup the new final assembly line to be ready for operation. During this period, operators can also be knowledgeable with new assembly process for both subassembly and final assembly lines. In addition, the new assembly line should be run as trail to observe the situation and perform adjustment if it's needed as well training operators on the new process. The new layout of subassembly line can be setup after the completion of first final assembly line, in case if production schedule is busy the new subassembly layout can be executed on the weekend to avoid production fallen off behind the schedule. After that, all operators should be trained on all subassembly and final assembly processes to be able to operate any station in case that the primary operator is out. Operators, who work or possibly will work to cover the master operator, should also be trained on new machines operation. In addition, work instructions for each

product family should be created to help operators understanding the process in training session as well using a reference for uncertainty assembly operation. Some operators may get frustrated because the new process operation times are balanced to each other and a team lead will cover an operator who wants to go to the restroom or get a drink to prevent stop production of fall behind. The existing final assembly line will be setup after completing the new assembly line to ensure having smooth transition and meet production schedule.

CONCLUSION

As a leading designer and manufacturer of beverage dispensers, Lancer Corporation must continuously strive to create and maintain a lean environment to uphold customer demands and expectations. The imbalanced cycle times between processes, especially the final assembly process 1 through testing 1 process caused a bottleneck in operation. Unavailability of work instructions for assemblies caused more operators needed to perform and observe a process. Utilizing lean principles and tools, the CTO System between the subassembly and final assembly was observed, mapped, evaluated, and improved. Value Stream Mapping is an important tool to utilize when trying to identify and eliminate waste. The resulting Future State Map offers a large reduction in unnecessary motion, inventory, waiting, eliminates batching, and adds innovative technology to the nozzle subassembly station to facilitate a more efficient process. The productivity rate, which was previously 20-24 units per day, was increased successfully to the desired 32 units per day to meet customer demands. Discussions with CTO manager Tatyana Sanchez are in progress to implement the Future State Map and increase production. Although implementation may prove difficult due to downtime and operator training requirements, this step towards a leaner environment will prove to be well worth the effort. In the words of Shigeo Shingo, a leading expert on manufacturing practices, “ The most dangerous kind of waste is the waste we do not recognize”;

value stream mapping allowed not only the recognition of waste within Lancer Corporation's CTO system, but also offered a tool to eliminate these wastes through the Future Stream Map. This will help Lancer Corporation to continue to deliver a high quality product and meet customer demands.

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